

Unit-1_MTH302_S25-26 (MCQs)

1. Discrete random variables and their probability distributions

8 If X is a discrete random variable, and its p.m.f is given by $f(x) = (x + 2)/25$, for $x = 1, 2, 3, 4, 5$, then $P(X \leq 4)$ is

(a) 18/25 (b) 2/25 (c) 10/25 (d) 6/25

2.

1. $\sum_{i=1}^n P(x_i)$ is equal to

(a) 0 (b) 1 (c) -1 (d) ∞

3.

A random variable X has the following probability function

X	:	0	1	2	3	4	5	6	7
$P(X)$	:	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

Find $P(1 < X < 5)$.

- A) 4/5
- B) 81/100
- C) 71/100
- D) 7/10

4.

Q: Determine the value c so that each of the following functions can serve as a probability distribution of the discrete random variable X :

$$f(x) = c \binom{2}{x} \binom{3}{3-x}, \text{ for } x = 0, 1, 2.$$

(a) 1/5 (b) 1/3 (c) 1/10 (d) 1

5.

Which of the following types of random variables can take on an infinite of possible values?

- (a) Discrete random variable
- (b) Continuous random variable
- (c) Both (a) and (b)
- (d) Neither (a) nor (b)

6.

Two dice are rolled. Let X denote the random variable which counts the total number of dots on the upturned faces. Then $F(4)$ is equal to

- a) $3/36$
- b) $4/36$
- c) $5/36$
- d) $6/36$

Answer: d) $6/36$

7.

Which of the following can be a probability mass function of a random variable?

- a) $f(x) = x/4$; $x = 0, 1, 2, 3$
- b) $f(1) = 0.5$, $f(2) = 0.6$, $f(-2) = -0.1$
- c) $f(x) = 1/4$; $x = -2, -1, 1, 2$
- d) $f(-1) = 0.33$, $f(1) = 0.33$, $f(0) = 0.33$

2. Continuous random variables and their probability distributions

8.

If a random variable X has a probability density function $f(x) = ax(2 - x)$, $0 < x < 2$, then the value of a is:

- (a) $3/4$
- (b) $1/4$
- (c) 2
- (d) 1

9.

2. If $f(x) = cx^2(1 - x)$, $0 \leq x \leq 1$ is a probability density function of a random variable X , then the value of c is

- (a) $1/12$
- (b) 6
- (c) 12
- (d) $1/6$

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If a random variable X has PDF $f(x) = 2x^2$, $0 < X < 1$, then $P(X < 0.5) = ?$

- A) 0.125**
- B) 0.25
- C) 0.5
- D) 0.75

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If $f(x) = kx^2$ for $0 \leq x \leq 2$, find k .

- A. $\frac{1}{2}$
- B. $\frac{3}{8}$
- C. $\frac{3}{4}$
- D. $\frac{1}{6}$

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The probability density function (PDF) of a continuous random variable X is given by

$$f(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{4}, & 0 \leq x < 2 \\ \frac{1}{4}, & 2 \leq x < 4 \\ 0, & x \geq 4 \end{cases}$$

Find the cumulative distribution function (CDF) $F(x)$.

$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{8}, & 0 \leq x < 2 \\ \frac{x-1}{4}, & 2 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$	$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{8}, & 0 \leq x < 2 \\ \frac{x}{4}, & 2 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$	$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{8}, & 0 \leq x < 2 \\ \frac{1}{2} + \frac{x-2}{4}, & 2 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$	$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{8}, & 0 \leq x < 2 \\ \frac{1}{2} + \frac{x-2}{2}, & 2 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$
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The continuous random variable has the following p.d.f.

$$f(x) = \begin{cases} \frac{x}{2}, & 0 < x \leq 1 \\ \frac{1}{4}(3-x), & 1 < x \leq 2 \\ \frac{1}{4}, & 2 < x \leq 3 \\ \frac{1}{4}(4-x), & 3 < x \leq 4 \\ 0, & o.w \end{cases}$$

What is $P(X > 4)$

- A) 0
- B) 1
- C) 0.5
- D) can not be determined

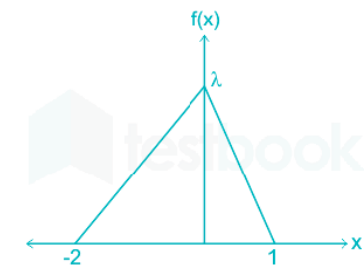
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$$f(x) = \begin{cases} \frac{1}{\beta - \alpha}, & \alpha \leq x \leq \beta \\ 0, & \text{otherwise} \end{cases}, \text{ if } \alpha = -1, \beta = 2, \text{ then } P\left(|X| \leq \frac{1}{2}\right) =$$

- (A) 1/3 (B) 1/2 (C) 5/6 (D) 2/3

15.

☐ Graph of $f(x)$ is shown below. For what value of λ , $f(x)$ can be used as pdf.



- (A) 2/3 (B) 3/4 (C) 4/5 (D) 1/2

3. Cumulative distribution functions

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The following is pdf of random variable X

$x:$	0	1	2	3	4	5	6	7
$p(x):$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

For $P(X \leq c) > \frac{1}{2}$, find the minimum value of c ,

- a) 2 b) 4 c) 5 d) 6

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If $F(x)$ is distribution function, then $F(-\infty)$ is

- (a) -1 (b) 0 (c) 1 (d) $-\infty$

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If the CDF of X is $F(x) = x^2$, $0 < x < 1$, then the probability that X lies between 0.2 and 0.6 is

A) 0.32

B) 0.40

C) 0.36

D) 0.10

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If $F(x)$ is a CDF, then $P(a < X \leq b) =$:

A. $F(a) + F(b)$

B. $F(b) - F(a)$

C. $F(b) \times F(a)$

D. $F(a)/F(b)$

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Q 1.2. Let X be random variable, $f(x)$ is p.d.f and $F(x)$ is c.d.f. Then what is relation between $f(x)$ and $F(x)$ for continuous distribution function.

(a) $F(x) = f'(x)$

(b) $F(x) = \int f(x)$

(c) $F'(x) = f(x)$

(d) There is no relation between them.

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A continuous random variable has the following c.d.f

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{2x^2}{5}, & 0 < x \leq 1 \\ \frac{3}{5} + \frac{2}{5} \left(3x - \frac{x^2}{2} \right), & 1 < x \leq 2 \\ 1, & x > 2 \end{cases}$$

Then $P(|x| > 1.5)$ is equal to

A) 0.75

B) 0.25

C) 0.36

D) 1

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☐ Let X be random variable with distribution function

$$F(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

Then $P(0 \leq 3X - 1 \leq 1) =$

(A) $3/5$

(B) $2/7$

(C) $1/3$

(D) 1

4. Moments about origin and mean (central moments) -Mean

23

A random variable X has the following probability distribution:

X	0	1	2
$P(X)$	0.3	0.5	0.2

What is the expected value $E(X)$?

- A) 0.5
- B) 0.9
- C) 1.0
- D) 1.5

24

If a random variable X has $E(X) = 2$ and $E(X^2) = 8$, then its variance is
(a) 4 (b) 6 (c) 8 (d) 2

25

$E(2X+3)$ is
(a) $E(2X)$ (b) $2E(X) + 3$ (c) $E(3)$ (d) $2X+3$

26

A fair coin is tossed 3 times. Let X be the number of heads. What is $E(X)$?

- a) 1
- b) 1.5
- c) 2
- d) 2.5

27

Q 1.3. For the discrete random variable X , the probability distribution is

x	0	1	2	3
$f(x)$	0.1	0.2	0.3	0.4

Then which of the following is correct

- (a) $E(X) = 1$
- (b) $E(X) = 2$
- (c) $E(X) = 3$
- (d) $E(X) = 4$

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The r -th moment about origin is defined as:

- A. $\mu_r = E[(X - \mu)^r]$
- B. $\mu'_r = E[X^r]$
- C. $E[X]$
- D. $E[(X - \bar{X})^r]$

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If the first moment about origin $\mu'_1 = 5$, then the mean of the distribution is:

- A. 0
- B. 2.5
- C. 5
- D. Undefined

30

A continuous random variable X has mean $\mu = 4$ and variance $\sigma^2 = 9$.

Find the second moment about the origin, $E(X^2)$.

- (a) 7
- (b) 13
- (c) 16
- (d) 25

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If a random variable's first moment is 5, what is its mean?

- a) 25
- b) 5
- c) 0.2
- d) None

32

A doctor recommended a patient to go on a particular diet for 2 weeks and there is equal likelihood for the patient to loose his weight between 2 kg and 4 kg. What is the average amount that patient is expected to loose on the diet.

- a) 2
- b) 3
- c) 4
- d) None

5. Moments about origin and mean (central moments)- Variance

33

Variance is always:

- a) Positive
- b) Negative
- c) **Zero or Positive**
- d) Zero or Negative

34.

11. Which of following is correct?

- (a) $\text{Var}(x) = E(x) - E(x^2)$
- (b) $\text{Var}(x) = E(x^2) - E(x)$
- (c) $\text{Var}(x) = E(x^2) - [E(x)]^2$**
- (d) $\text{Var}(x) = [E(x^2)]^2 - E(x^2)$

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67. $\text{Var}(aX+b)$

- (a) $ab \text{Var}(X)$
- (b) $\frac{a}{b} \text{Var}(X)$
- (c) **$a^2 \text{Var}(X)$**
- (d) $a \text{Var}(X)$

36

$\text{Var}(X+8)$ is

- (a) $\text{var}(8)$
- (b) $\text{var}(X)$
- (c) $8 \text{var}(X)$
- (d) 0

37

If $E[X] = 10$, then the first central moment is:

- A. 10
- B. 0
- C. 1
- D. Undefined

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If the mean of a distribution is 4 and variance is 9, then the second moment about origin is:

- A. 25
- B. 13
- C. 5
- D. 10

39

If the third moment about mean is zero, then the distribution is:

- 1. Positively skewed
- 2. Negatively skewed
- 3. Symmetrical
- 4. Mesokurtic

40

A continuous random variable has p.d.f. given by

$$f(x) = \begin{cases} a^2 x e^{-ax}, & x \geq 0 \\ 0, & \text{o.w} \end{cases}$$

Then $V(x)$ is equal to

a) $\frac{2}{a^2}$

b) $\frac{2}{a}$

c) $\frac{2}{a^3}$

d) $\frac{6-2a}{a^2}$

6. Moments about origin and mean (central moments)- Skewness

41

2. For a given dataset, the first raw moment about the origin is 10, and the second central moment is

25. What is the value of the second raw moment about the origin?

- a) 25
- b) 100
- c) 125
- d) 35

42

For a negatively skewed distribution, the correct inequality is

- (a) Mode < median
- (b) mean < median
- (c) mean > median
- (d) None of the above

43

If mean = 60, median = 55, and standard deviation = 10, then Karl Pearson's coefficient of skewness = ?

- a) 0.5
- b) 1.5

- c) 0.3
- d) 1.0

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A **positively skewed distribution** has:

- A. Tail towards the left
- B. Tail towards the right
- C. Equal tails
- D. No tails

45

For a **perfectly symmetrical distribution**, the mean, median, and mode are:

- A. All equal
- B. All unequal
- C. Mean > Median > Mode
- D. Mean < Median < Mode

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The first four moments about the origin of a random variable X are given as:

$$\mu'_1 = 2, \quad \mu'_2 = 10, \quad \mu'_3 = 40, \quad \mu'_4 = 170.$$

Find the **second, third, and fourth central moments** (μ_2, μ_3, μ_4).

- A) $\mu_2 = 6, \mu_3 = 8, \mu_4 = 26$
- B) $\mu_2 = 6, \mu_3 = 8, \mu_4 = 42$
- C) $\mu_2 = 8, \mu_3 = 6, \mu_4 = 42$
- D) $\mu_2 = 6, \mu_3 = 0, \mu_4 = 36$

7. Moments about origin and mean (central moments)- kurtosis

47

1. The fourth central moment (μ_4) is most directly used in the calculation of which statistical measure?

- a) Variance
- b) Skewness
- c) Kurtosis
- d) Mean

48

If $\beta_2 = 3$, the distribution is called

- (a) Mesokurtic (b) Leptokurtic (c) Platykurtic (d) None of the above

49

Which of the following best describes Leptokurtic distributions?

- a) Light tails, flat top
- b) Heavy tails, sharp peak
- c) Symmetric with low variance
- d) Skewed to the left

50

Q 1.7. *The fourth moment(Kurtosis) predict about*

- (a) *Central location of data*
- (b) *Spread/Dispersion around the mean*
- (c) *Direction and degree of asymmetry*
- (d) *Shape (peaked vs flat)*

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The moment coefficient of kurtosis is defined as:

- A. $\beta_1 = \frac{\mu_3^2}{\mu_2^3}$
- B. $\beta_2 = \frac{\mu_4}{\mu_2^2}$
- C. $\beta_2 = \frac{\mu_2}{\mu_4}$
- D. $\beta_1 = \frac{\mu_2}{\mu_3^2}$

52

For a **Leptokurtic** distribution, the coefficient of kurtosis β_2 :

- A. < 3
- B. $= 3$
- C. > 3
- D. $= 0$

53

For a random variable X , the following **central moments** are given:

$$\mu_2 = 5, \quad \mu_3 = 4, \quad \mu_4 = 80$$

Find the **coefficients of skewness** and **kurtosis** of the distribution.

- A) Skewness = 0.36, Kurtosis = 3.2
- B) Skewness = 0.40, Kurtosis = 3.2
- C) Skewness = 0.20, Kurtosis = 2.5
- D) Skewness = 0.50, Kurtosis = 3.0

54 & 55

The central moments of a distribution are given by $\mu_2 = 140, \mu_3 = 148, \mu_4 = 6030$, the value of moment coefficient of skewness (β_1) is-

- (a) 0.21 (b) 0.0079 (c) 0.0095 (d) None of these

Ans B

For a distribution the mean is 10, variance is 16 and moment coefficient of Kurtosis $\beta_2 = 4$. The value of fourth moment μ_4 is –

- (a) 1011 (b) 950 (c) 1024 (d) 2025

Ans C

56

If X is a discrete random variable assuming the values 0, 1, 2, 3, 4 and $F(x)$ is its cumulative probability distribution, then $P(1 < X < 3)$ is

- (a) $F(3) - F(1)$
- (b) $F(3) - F(2)$
- (c) $F(1) + F(3)$
- (d) $F(1) - F(3)$

57

Cumulative distribution function is related to

- (a) Discrete Random variable
- (b) Continuous Random Variable
- (c) Both Discrete and Continuous
- (d) None

58

If X is a random variable with probability density function $f(x) = kx$; $0 < x < 1$, then the value of k is

- (a) 1
- (b) 4
- (c) $\frac{1}{2}$
- (d) 2

59

If X is a random variable having the probability density function $f(x) = (1/3)e^{-x/3}$; $x > 0$ then $P(X > 3)$ is

- (a) $1/e$
- (b) 1
- (c) $\frac{1}{3}$
- (d) 0.75

60

Let X be the random variable with probability distribution as

x	0	1	2	3
$P(X=x)$	$\frac{1}{3}$	$\frac{1}{2}$	0	$\frac{1}{6}$

Then the Mean is

- (a) 1
- (b) 2

- (c) 1.5
- (d) 3

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Let X be the random variable with probability distribution as

x	0	1	2	3
P(X=x)	$\frac{1}{3}$	$\frac{1}{2}$	0	$\frac{1}{6}$

Then the Variance is

- (a) 1
- (b) 2
- (c) 1.5
- (d) 3

62

If X is a random variable with probability density function $f(x) = 2x$; $0 < x < 1$, then the variance is

- (a) 1
- (b) $\frac{4}{18}$
- (c) $\frac{1}{2}$
- (d) $\frac{1}{18}$